Dasaroju Vineethkumar

MTech
Design and Manufacturing
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech.	Indian Institute of Technology, Tirupati	8.23	2024-2026
B.Tech. (ME)	JNTUH University College of Engineering, Sultanpur	7.48	2021-2024
Diploma (ME)	Jawaharlal Nehru Govt. Polytechnic	9.71	2018-2021
Secondary	TG BSE Board	9.7	2018

Experience

- **ST Motors** *June 2020 - Dec. 2020*
  Intern-Mechanic Hyderabad
  – Utilized company equipment to perform diagnostics, repair, and research issues on motorcycles and equipment.Maintained and repaired motorcycle parts, including engines, brakes, and fuel/exhaust systems.

Projects

– **Design and Fabrication of Indigenous piezoelectric accelerometer** *Present.*
Currently working on the design of a piezoelectric accelerometer for vibration measurement in dynamic systems.
∗ **Tools & technologies used**: ANSYS workbench, Arduino IDE, COMSOL Multiphysics
∗ Focused on material selection, sensitivity optimization, and frequency response analysis through simulation models. Designed for application in machine dynamics with an emphasis on low-cost, high-sensitivity sensor design.

– **Design and Optimization of the disc brake rotor** *Jan. 2024 - May 2024*
This project aims to develop a new disc rotor profile to improve heat dissipation and structural strength.
∗ **Tools & technologies used**: Creo PTC, ANSYS workbench
∗ Results showed that the new profile offered improved heat dissipation and reduced thermal distortion, leading to enhanced braking efficiency and durability.

– **Design and Optimization of the Mechanical Rover** *June 2023*
This project aims to develop and improve the durability and performance of a multi-terrain mechanical rover.
∗ **Tools & technologies used**: Creo PTC, ANSYS workbench
∗ Modeled 3D geometry in Creo and carried out static and dynamic analysis in ANSYS.

– **Design and Fabrication of solar powered hacksaw machine** *Jan.2021 - May.2021*
Designed and developed a solar powered hacksaw for light-duty metal cutting using a crank-slider mechanism.
∗ **Tools & technologies used**: Auto CAD, SolidWorks, ANSYS workbench
∗ Modeled all mechanical components in SolidWorks and optimized for manufacturability and performance.Focused on eco-friendly design principles and energy efficiency for small-scale workshop applications.

Technical Skills

–**Programming:** Python, MATLAB
–**Design & CAD Tools:** AutoCAD, Creo PTC, SolidWorks
–**Libraries/Frameworks:** Pandas, Numpy, scikit-learn
–**Workshop Engineering:** Carpentry, Welding, Foundry, Machine shop
–**Simulation & Analysis:** ANSYS workbench, COMSOL Multiphysics, Simulink
–**Productivity Tools:** Microsoft Office Suite, Google Workspace

Areas of Interest

–Computer-Aided Engineering (CAE), Advanced Manufacturing Technologies, Structural and Mechanical System Design, Kinematics and Dynamics of Machines

Key Courses Taken

∗Advanced Mechanics of Solids, Applied Numerical Methods, Finite Element Methods, Product Design and Development, Design for Manufacturing and Assembly, Vibrations of Discrete Systems

Positions of Responsibility

∗ **Head Coordinator,** Technical club, JNTUH UCES *Apr. 2023 - May 2023*

∗ **Techinal Fest Coordinator,** Mecchorenza, JNTUH UCES *Apr. 2022 - May 2022*

Certifications

∗ Certification on Automobile Servicing and Maintenance

Extracurricular Activities

∗**Athletics:** Active track runner, participated in inter-college running competitions.
∗**Event Volunteer:** Assisted in organizing workshops during college technical festival.